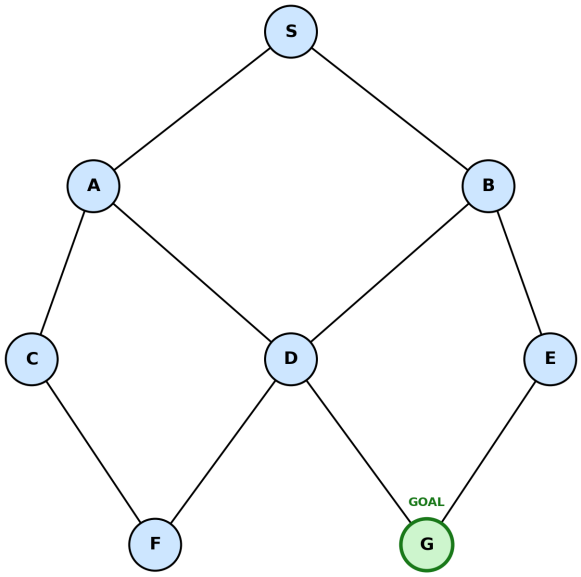
Artificial Intelligence (Practice Problem Set)

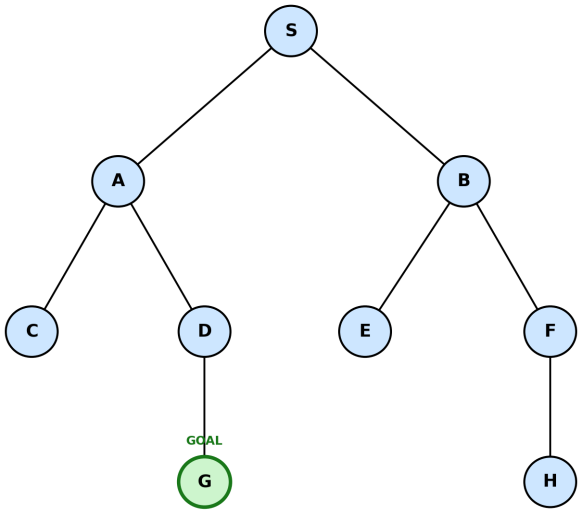
Q1. Consider the undirected graph below with start node S and goal node G.



i. Show the order in which nodes are visited and expanded by BFS and DFS, starting from S, exploring neighbours in alphabetical order.

ii. Compare the path found by DFS with the path found by BFS. Is DFS's path optimal? Why or why not?

Q2. Consider the search tree below, with start node S and goal node G at depth 3.

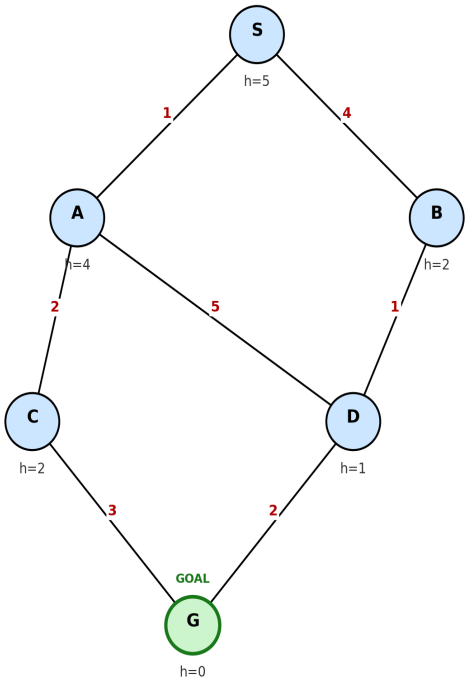


(a) Perform IDS starting with depth limit L = 0, increasing L by 1 each iteration. For EACH value of L (0, 1, 2, 3), list the nodes expanded during that depth-limited DFS iteration.

(b) At which depth limit L is the goal G found? Give the resulting path from S to G.

(c) IDS re-expands many of the shallow nodes multiple times across iterations. Explain briefly why IDS is still considered time-efficient despite this repeated work.

Q3. Consider the weighted graph below (edge labels are path costs) having heuristic values at each node, with start node S and goal node G.



(a) Perform UCS from S to G. Show the frontier after each expansion. State the path found and its total cost.

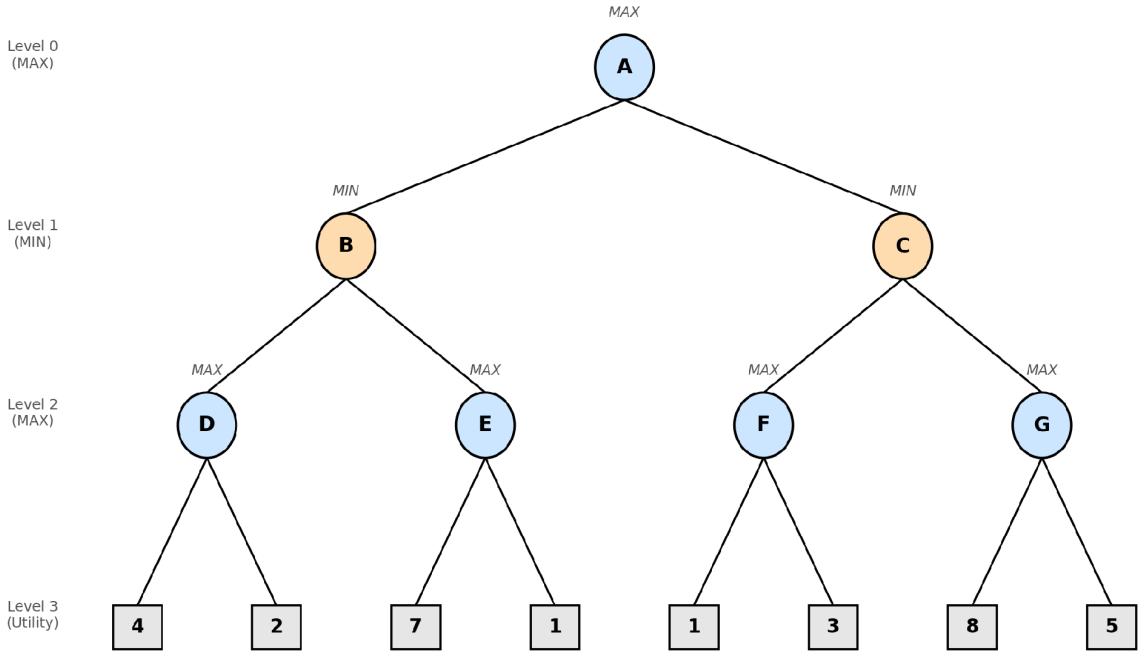
(b) Perform Greedy Best-First Search from S to G, expanding at each step the frontier node with the LOWEST heuristic value h(n) only

(c) Perform A* search from S to G, expanding at each step the frontier node with the lowest f(n) = g(n) + h(n). State the path found and its total cost.

(d) Perform IDA*: show which nodes are expanded, which are pruned (because f(n) exceeds the limit), and determine the new f-cost limit for each iteration till the goal is found.

(e) Perform RBFS execution step by step and the final path and cost found from start to goal.

Q4. Consider the game tree below. Node A is the root (MAX to move). The tree alternates MAX/MIN by level, and the Level-3 values are the terminal utility values. Assume children are explored strictly LEFT to RIGHT.



(a) Compute the plain Minimax value of every internal node (D, E, F, G, B, C, A), without any pruning.

(b) Now re-run the search using Alpha-Beta pruning on the same tree. At every internal node, show how α and β are updated as each child returns its value.

(c) True or False: "Alpha-Beta pruning can, in some cases, find a different (better or worse) final root value than plain Minimax on the same tree." Justify your answer.